

Unit 6, Lesson 6: Using Two-Column Tables of Equivalent Ratios

Benchmarks

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

Learning Targets

- ☐ I can use two-column tables to display equivalent ratios.
- ☐ I can solve problems using two-column tables of equivalent ratios.

6.6.1 Warm-Up: Different Ways of Expressing Division

In many Latin American countries and other countries around the world, the colon (:) is used instead of $\div$ to express division. Therefore, the expression “28 divided by 4” would be expressed as 28:4 instead of $28 \div 4$.

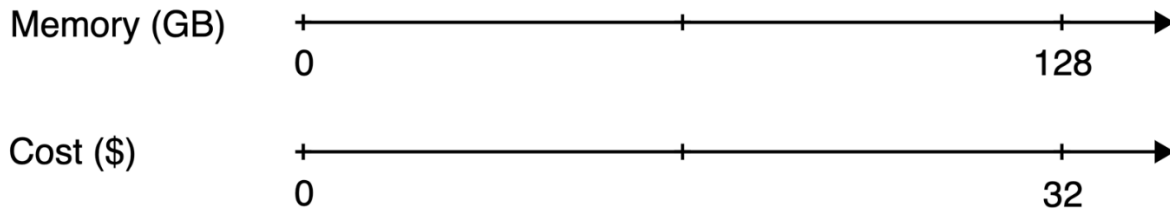
1. Using what you have learned about ratios, explain how 28:4 and $28 \div 4$ can be used to express the same value.
2. Write one question that comes to your mind after reading the information shown.

6.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Work with your partner on the following.

In 2016, 128 gigabytes (GB) of portable computer memory cost \$32.

A double number line diagram that represents the situation is shown.



1. One set of tick marks has already been drawn to show the result of multiplying 128 and 32 each by $\frac{1}{2}$. Label the amount of memory and the cost for these tick marks.
2. Next, keep multiplying by $\frac{1}{2}$ and drawing and labeling new tick marks until you can no longer clearly label each new tick mark with a number.
3. Discuss with your partner whether or not it's possible to clearly represent the cost of 1 GB of memory on the double number line diagram.
4. If it is possible, circle the cost of 1 GB of memory on the double line diagram. If it is not possible, explain why not.

5. A table representing the same situation is shown. Work with your partner to find the cost of 1 GB. Use as many or few rows as needed.

Memory (gigabytes)	Cost (dollars)
128	32

6. Explain your steps for determining the cost per gigabyte using the table.
7. Use either the double line diagram or the ratio table to determine the cost of 512 GB of portable computer memory at this rate.
8. Discuss with your partner which representation of the equivalent ratios you preferred using to find the costs of 1 GB and 512 GB. Summarize your discussion.



The IBM 7094 was regarded as one of the most powerful advanced mainframe computers in the early 1960s. NASA and the Air Force used it in the Gemini and Apollo space programs.

Despite its large size and historical impact, the IBM 7094 only had a core memory of about 150 kilobytes (KB). That's only enough to manage a few Microsoft Word documents.



Today, you can get a small USB flash drive like the one shown for less than \$20 that can store 4 gigabytes (GB) of memory.



That's almost 27,000 times the memory as the IBM 7094, and you can fit it in your pocket!

6.6.3 Guided Instruction: Using Two-Column Ratio Tables

Both double number lines and tables can be useful tools for finding equivalent ratios, but there is an important distinction between the two representations. On a double number line, the numbers on each line are listed in order. On a double number line, the numbers on each line are listed in order.

With a table, ratios can be written in any order. For this reason, sometimes a table is easier to use to solve a problem.

Lin is paid \$90 for 5 hours of work. She used the following table to calculate how much she would be paid at this rate for 8 hours of work.

	Amount Earned (\$)	Time Worked (hours)	
$\frac{1}{5} \times$	90	5	$\times \frac{1}{5}$
	18	1	
$8 \times$	144	8	$\times 8$

1. What is the meaning of the ratio that appears in the second row of the table, 18:1?
2. Why was the number $\frac{1}{5}$ used as a factor?

3. How else could the values in the second row be determined from the first row?
4. Complete the statements to explain how Lin used this table to solve the problem.

Lin used equivalent ratios to solve the problem. Lin first determined the rate for

- ☐ 1 hour
- ☐ 5 hours
- ☐ 8 hours

of work by multiplying each value in the given ratio by $\frac{1}{5}$. She

then used that unit rate to find how much she would be paid for 8 hours by

_____.

5. At this rate, how much would Lin be paid for 2.25 hours of work?

6.6.4 Guided Practice: Using Two-Column Ratio Tables

1. Rosalie runs $\frac{1}{3}$ of a mile in 4 minutes. As part of her marathon training, her goal today is to run 7.5 miles at this rate. Use the table to determine how long it will take her to run 7.5 miles at this rate. Show how the values in each row are determined using arrows and multiplication.

Distance (miles)	Time (minutes)
$\frac{1}{3}$	4

6.6.5 Your Turn!

1. The International Space Station (ISS) orbits around the Earth at a constant speed.

The ISS completes 2 orbits around the Earth in approximately 3 hours. Complete the table to find the approximate number of times the ISS travels around the Earth in 1 hour and in 1 day (24 hours).



Orbits Around Earth Completed	2		
Time (hours)	3	1	24

2. At a local natural food store, customers can measure and purchase foods in any quantity. Patricia purchased 4 pounds of almonds for \$38.48. Greg only needs to purchase 1.5 pounds of almonds. Use the table to determine the cost for 1.5 pounds of almonds at the natural food store.

Almonds (pounds)	Cost (dollars)
4	38.48

6.6.6 Wrap-Up: Ticket Out the Door

1. Julio participated in a walk-a-thon to raise money for cancer research. Walking at a constant speed, Julio traveled 6.4 miles in the first 2 hours of the walk-a-thon. His goal is to walk for 7 hours. Use the table to find how far Julio will have walked if he reaches his 7-hour goal. Show how the values in each row are determined.

Hours	Miles Walked
2	6.4